Statistical Testing on E-Commerce Data — Report (Part I)

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Dataset description

The analysis uses the Online Retail "E-Commerce Data" dataset from Kaggle, containing invoice-level records of a UK-based online retailer, including product description, quantity, unit price, invoice date, customer ID, and country. After restricting to rows with strictly positive `quantity` and `unit_price` (excluding returns/cancellations and data-entry errors) and constructing `revenue = quantity × unit_price`, the working dataset contains **530,104 rows** across **38 distinct countries** and **4,338 distinct customers**. The raw file held 541,909 rows; the 11,805 dropped rows were returns, cancellations, or zero-price adjustments.

`quantity`, `unit_price`, and the derived `revenue` are all strongly right-skewed, with a small number of very large bulk orders and high-unit-price items pulling the mean well above the median. For revenue the median is £9.90 while the mean is £20.12; the maximum revenue on a single line item is £168,469.60 (a bulk order of 80,995 units). The data are dominated by the **United Kingdom**, which accounts for **91.5%** of transactions (485,123 of 530,104 line items). The next largest countries by transaction count are Germany (9,040), France (8,407), EIRE (7,890), and Spain (2,484).

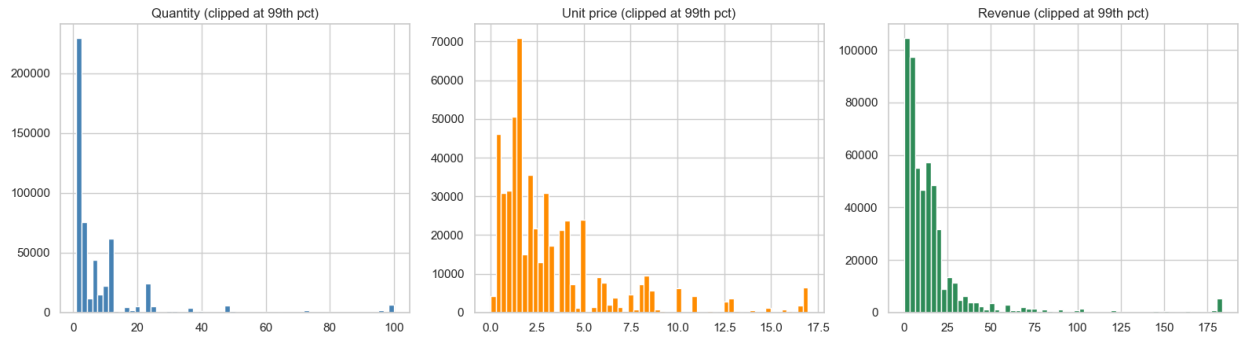


Figure 1: Distributions

Figure 1. Distributions of quantity, unit price, and revenue (clipped at the 99th percentile for visualisation). All three are strongly right-skewed with a long upper tail.

Test 1: Normality of revenue

Question: does the distribution of line-item revenue resemble a normal distribution?

Test used: Shapiro-Wilk test on a random sample of 3,000 revenue values.

Hypotheses: - H0: the sampled revenue values come from a normal distribution. - H1: the sampled revenue values do not come from a normal distribution.

Result: $W = 0.3038$, $p \approx 5.97 \times 10^{-75}$.

Interpretation. At $\alpha = 0.05$ we **reject H0**: the sampled revenue values do not come from a normal distribution. This is exactly as expected — revenue is a strictly positive quantity formed as the product of two

positive, right-skewed variables, so its distribution is heavily right-skewed with a long upper tail, as the histogram confirms. The practical implication is that later tests which formally assume approximate normality of the response within groups (the two-sample t-test in Test 3 and the one-way ANOVA in Test 4) are applied to a non-normal response. They are reasonably robust to this violation at the very large sample sizes here (hundreds of thousands of observations per group), but the result should still be interpreted with some caution, and it motivates the non-parametric Mann-Whitney U test added as Test 6. The non-parametric tests (Spearman in Test 2, chi-square in Test 5) do not rely on a normality assumption at all.

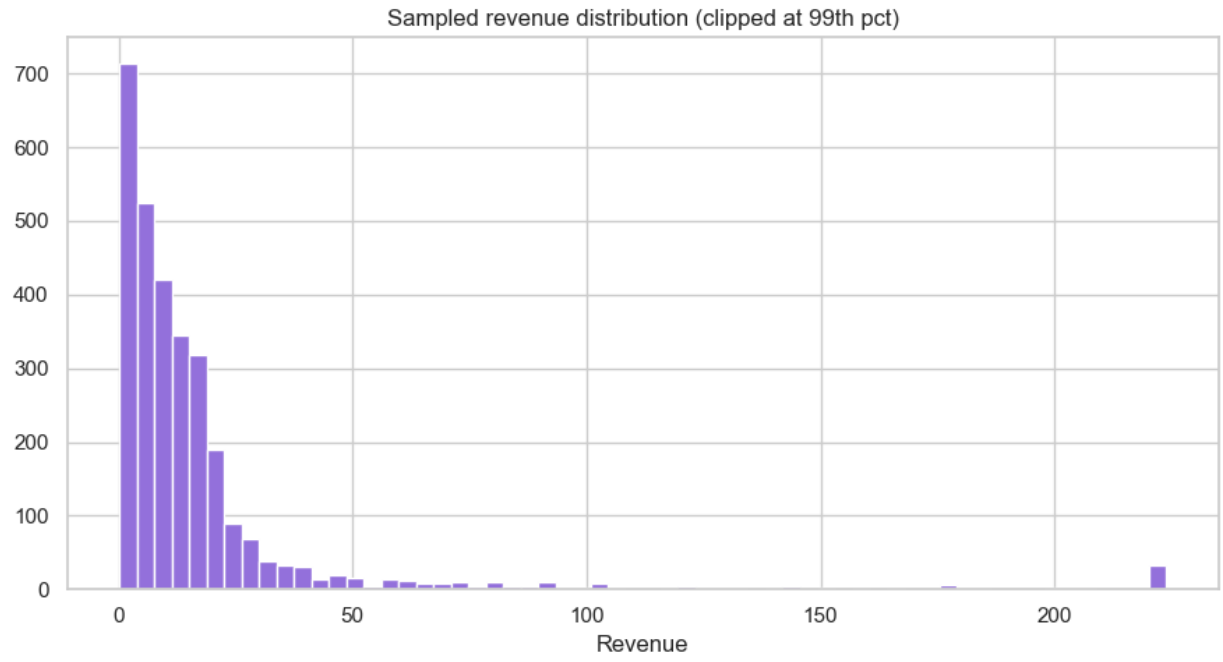


Figure 2: Revenue sample histogram

Figure 2. Sampled revenue distribution (3,000-observation sample, clipped at 99th percentile). The strong right skew is clearly visible.

Test 2: Association between quantity and unit price

Question: is there an association between quantity ordered and unit price?

Test used: Spearman rank correlation.

Hypotheses: - H0: there is no monotonic association between quantity and unit price. - H1: there is a monotonic association between quantity and unit price.

Result: $r_s = -0.4032$, $p \approx 0$.

Interpretation. We **reject H0**: there is a statistically significant monotonic association between quantity and unit price. The sign is **negative**, meaning that customers tend to buy larger quantities of cheaper items and smaller quantities of more expensive items — consistent with bulk-buying of low-cost goods and smaller basket sizes for high-unit-price items. In terms of magnitude, $|r_s| = 0.40$ is a **moderate** association (not weak, but far from the near-perfect monotone relationships seen in physical data). With ~530,000 observations the test has enormous power, so statistical significance is essentially guaranteed; the more meaningful

observation is that the effect size is moderate rather than strong, suggesting pricing and purchase quantity are related but not deterministically so.

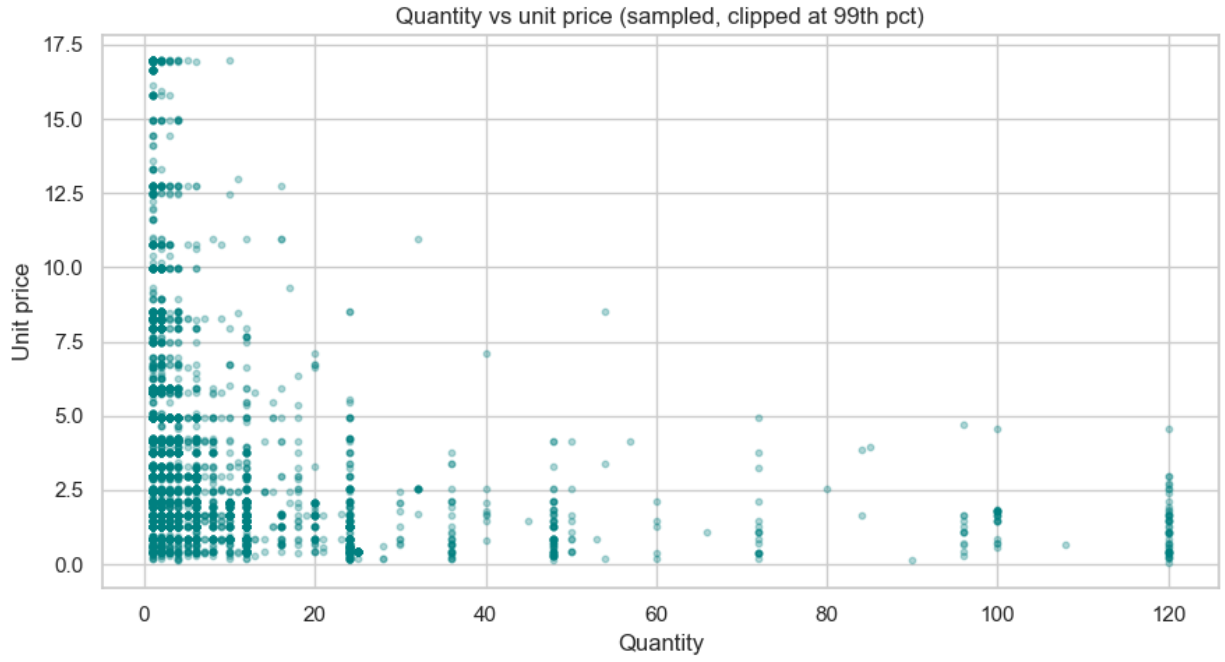


Figure 3: Quantity vs price

Figure 3. Quantity vs unit price (5,000-observation sample, clipped at 99th percentile). The negative association is visible as a denser cloud of high-quantity / low-price points.

Test 3: Revenue difference between the United Kingdom and other countries

Question: do line-item revenues differ on average between the UK and all other countries combined?

Test used: Welch's two-sample t-test (unequal variances assumed, given very different group sizes).

Hypotheses: - H0: mean revenue for UK transactions equals mean revenue for non-UK transactions. - H1: mean revenue for UK transactions differs from mean revenue for non-UK transactions.

Result: UK mean = £18.60 (n = 485,123, std = 281.28); other countries mean = £36.49 (n = 44,981, std = 88.38); $t = -30.83$, $p \approx 5.0 \times 10^{-208}$.

Interpretation. The **non-UK** group has the higher mean revenue per line item: £36.49 versus £18.60 for the UK. We **reject H0** at $\alpha = 0.05$ — the difference in mean line-item revenue is statistically significant. The direction makes commercial sense: the retailer is UK-based, so UK transactions are dominated by high volume of smaller everyday orders, while non-UK transactions tend to be larger wholesale-style line items. Because the UK sample is so large, even a modest absolute difference is highly significant; the £17.89 gap in means is also practically meaningful in currency terms — non-UK line items are on average nearly twice as valuable as UK ones. We note that the very large standard deviations relative to the means reflect the extreme right skew established in Test 1, which is why the non-parametric Test 6 is the more robust comparison.

Test 4: Revenue across top countries

Question: among the five countries with the most transactions, are the mean revenues the same?

Test used: one-way ANOVA (`scipy.stats.f_oneway`) across the top 5 countries by transaction count.

Hypotheses: - H0: mean revenue is equal across all five top countries. - H1: at least one country's mean revenue differs from the others.

Result: top 5 countries = United Kingdom, Germany, France, EIRE, Spain; $F = 10.22$, $p \approx 2.83 \times 10^{-8}$.

Country	Mean revenue (£)
United Kingdom	18.60
Germany	25.32
France	24.95
EIRE	35.93
Spain	24.79

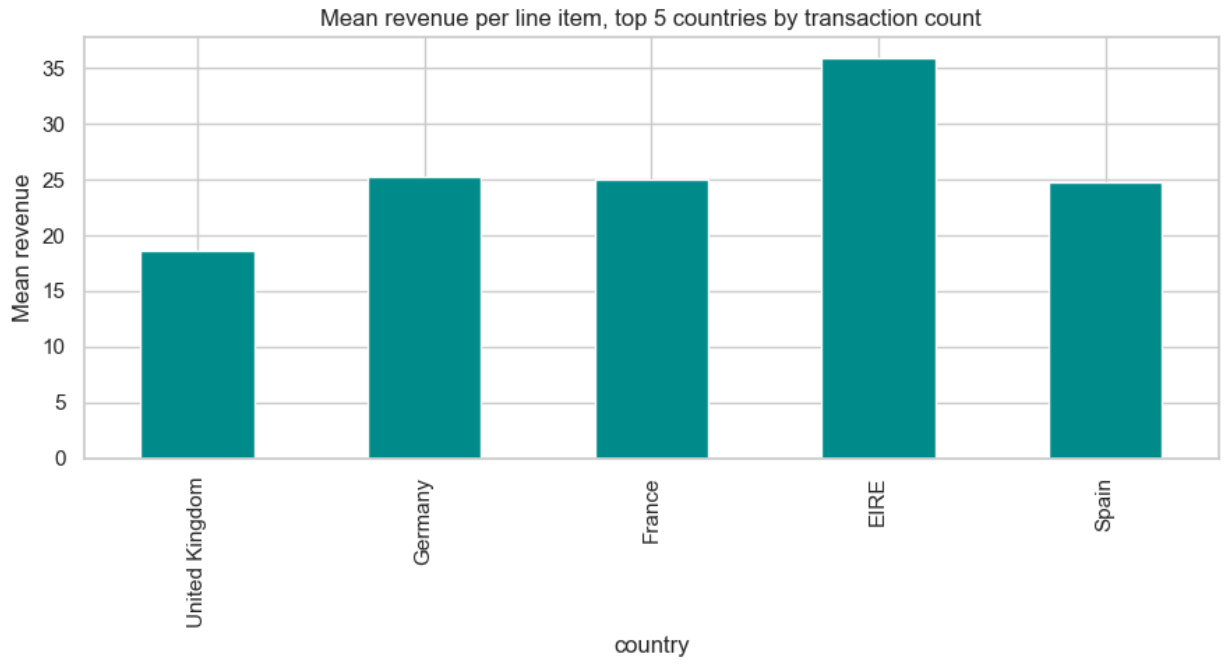


Figure 4: ANOVA means

Figure 4. Mean revenue per line item for the top 5 countries by transaction count. EIRE is the clear outlier on the high side; the UK is lowest.

Interpretation. We **reject H0** at $\alpha = 0.05$: at least one country's mean line-item revenue differs from the others. **EIRE has by far the highest mean revenue (£35.93)**, well above Germany (£25.32), France (£24.95), Spain (£24.79), and the UK (£18.60). The UK's low mean reflects the same volume-of-small-orders pattern noted in Test 3, while EIRE's high mean points to large wholesale-style orders. A significant ANOVA only establishes that *some* difference exists; it does not identify which specific pair(s) differ. A post-hoc Tukey

HSD test would be the natural follow-up to make pairwise comparisons while controlling the family-wise error rate.

Test 5: Association between country and high-value purchase

Question: is the proportion of high-value purchases the same across the main countries?

Test used: chi-square test of independence on a country × high-value contingency table. "High value" is defined as revenue at or above the 90th percentile of the filtered dataset (£32.85); this is a stated modelling choice.

Hypotheses: - H0: high-value purchase status is independent of country. - H1: high-value purchase status is associated with country.

Result: $\chi^2 = 2489.52$, df = 4, $p \approx 0$.

Observed contingency table:

Country	Not high value	High value
United Kingdom	442,333	42,790
Germany	7,466	1,574
France	7,050	1,357
EIRE	6,271	1,619
Spain	2,226	258

Expected counts under H0:

Country	Not high value	High value
United Kingdom	440,106.6	45,016.4
Germany	8,201.1	838.9
France	7,626.9	780.1
EIRE	7,157.9	732.1
Spain	2,253.5	230.5

Figure 5. Share of high-value line items by country (top 5 by transaction count). EIRE is highest at 20.5%; the UK is lowest at 8.8%.

Interpretation. We **reject H0** at $\alpha = 0.05$: high-value purchase status is strongly associated with country. The high-value shares are:

Country	High-value share (%)
EIRE	20.52
Germany	17.41
France	16.14
Spain	10.39
United Kingdom	8.82

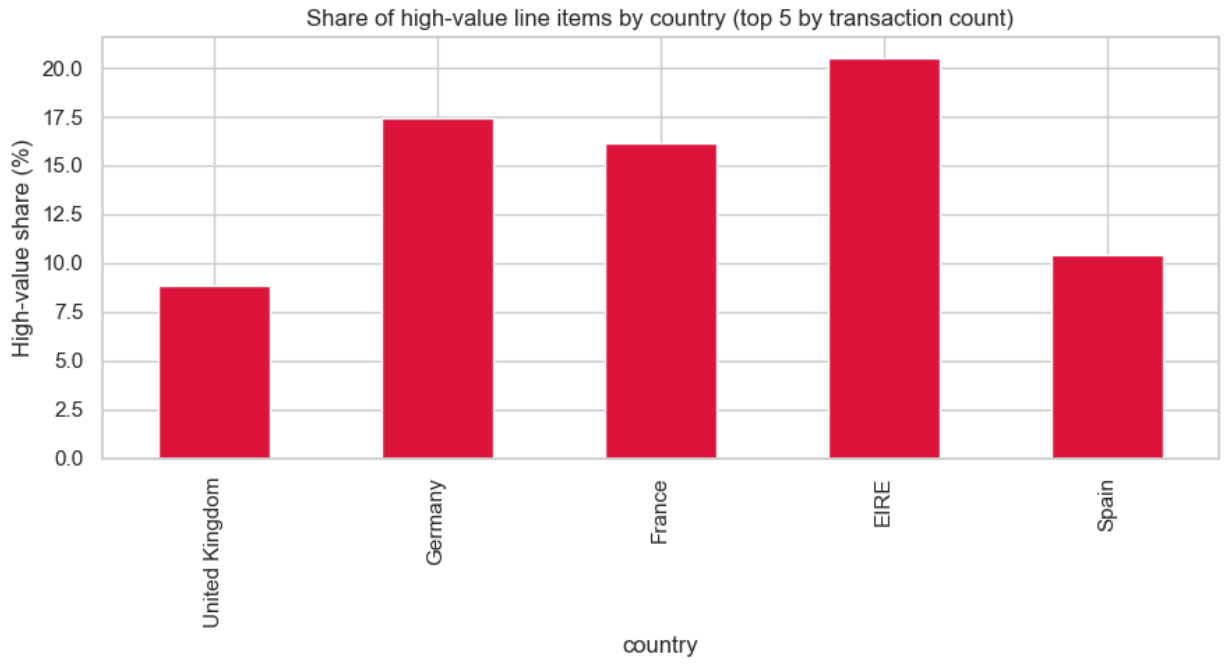


Figure 5: High-value rate

EIRE has the highest share of high-value line items (20.5%, well above the ~10% expected under independence), followed by Germany (17.4%) and France (16.1%). The **United Kingdom** has the lowest share (8.8%, *below* the ~10% expected), confirming the pattern from Tests 3 and 4: UK transactions are numerous but individually small, while the other top countries carry a disproportionately large share of high-value line items. The observed-vs-expected counts show the UK has ~2,226 fewer high-value items than expected under independence, while EIRE has ~887 more — a large and consistent deviation.

Test 6: Non-parametric comparison of UK vs other countries (Mann-Whitney U)

Question: given that Test 1 established revenue is not normally distributed, do UK and non-UK line-item revenues differ in their *distributions* (not just their means) using a non-parametric test?

Test used: Mann-Whitney U (Wilcoxon rank-sum), the non-parametric counterpart to Welch's t-test.

Hypotheses: - H0: the distribution of revenue is the same for UK and non-UK line items (equal ranks/medians). - H1: the distribution of revenue differs between UK and non-UK line items.

Result: $U = 5{,}712{,}161{,}120$, $p \approx 0$. UK median = £8.42; non-UK median = £17.40.

Interpretation. We **reject H0** at $\alpha = 0.05$: the distribution of revenue differs between UK and non-UK line items. The medians tell the story more robustly than the means for this right-skewed data: the UK median revenue is £8.42 while the non-UK median is £17.40 — roughly double. This **agrees with Test 3's conclusion** (non-UK revenues are higher), but the Mann-Whitney test compares the full distributions via ranks rather than just the means, so it is the more defensible comparison given the strong non-normality established in Test 1. The fact that both the parametric (t-test) and non-parametric (Mann-Whitney) tests agree gives confidence that the UK-vs-rest difference is a real feature of the data, not an artifact of the normality violation.

Summary of results

#	Test	Question	Statistic	p-value	Decision at α=0.05
1	Shapiro-Wilk	Is revenue normally distributed?	$W = 0.3038$	$\sim 5.97 \times 10^{-75}$	Reject H0
2	Spearman	Quantity vs unit price association?	$r_s = -0.4032$	~0	Reject H0
3	Welch's t-test	UK vs other countries, mean revenue?	$t = -30.83$	$\sim 5.0 \times 10^{-208}$	Reject H0
4	One-way ANOVA	Mean revenue across top 5 countries?	$F = 10.22$	$\sim 2.83 \times 10^{-8}$	Reject H0
5	Chi-square	High-value purchase vs country?	$\chi^2 = 2489.52$, df=4	~0	Reject H0
6	Mann-Whitney U	UK vs other, revenue distribution?	$U = 5.71 \times 10^9$	~0	Reject H0

Closing reflection

All six null hypotheses are rejected at $\alpha = 0.05$. The most appropriate tests for this dataset are the non-parametric ones: **Spearman** (Test 2), **chi-square** (Test 5), and **Mann-Whitney U** (Test 6). Because Test 1 firmly established that revenue is strongly non-normal ($W = 0.30$, $p \approx 10^{-75}$), the t-test (Test 3) and ANOVA (Test 4) rest on a violated normality assumption. They remain broadly trustworthy here only because of the very large per-group sample sizes (tens to hundreds of thousands of observations), but the non-parametric alternatives are the more defensible choices. Notably, the Mann-Whitney test (Test 6) and Welch's t-test (Test 3) agree on the UK-vs-rest conclusion, which gives confidence the finding is genuine rather than an artifact of the non-normality.

The most questionable assumption across all tests is the **independence of observations**. The dataset is at the line-item level, but multiple line items belong to the same invoice and the same customer — these are not independent draws. This inflates effective sample sizes and makes small differences statistically significant more readily than they would be under true independence (e.g. the t-test's p-value of $\sim 10^{-208}$). A more rigorous follow-up would aggregate to the customer or invoice level before re-running the between-country comparisons, which would reduce the sample size but yield more honest p-values. A second limitation is the extreme imbalance toward the UK (91.5% of rows), which means non-UK conclusions rest on a comparatively small subset and the "UK vs other" comparisons are dominated by a handful of countries (EIRE, Germany, France) whose commercial relationship to the retailer may differ.

References

1. Kaggle. *Online Retail "E-Commerce Data."* https://www.kaggle.com/datasets/carrie1/ecommerce-data